

QUICK START INSTRUCTIONS

Routine maintenance
Heater-Cooler System 3T

TIMELINES FOR DISINFECTION AND RELATED TASKS

- A) prior to initial operation;
prior to storing

→

Surface disinfection and
disinfection of water circuits

page 2
- B) after every operation

→

Surface disinfection

page 2
- C) every 7 days

→

Water change;
Add hydrogen-peroxide;
Disinfection of overflow bottle

page 3
page 3
page 4
- D) every 14 days

→

Disinfection of water circuits

page 5
- E) once per year

→

Exchange the tubings

page 10
- F) regularly

→

Take water samples

page 10

NOTE: These Quick Start Instructions are only an excerpt of the Operating Instructions (CP_IFU_16-XX-XX_USA_014) regarding routine water maintenance and disinfection (chapter 5 and 6). They do not replace the Operating Instructions. For complete set up, use of the machine and warnings and cautions, the comprehensive Heater Cooler System Operating Instructions must be followed.

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Heater-Cooler System 3T

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A) prior to initial operation; prior to storing

QUICK START INSTRUCTIONS | Heater-Cooler System 3T

A.1) SURFACE DISINFECTION

1.1

Close the CAN-jack with the matching cover.

1.2

Use only pre-soaked, ready-to-use disinfectant tissues.

1.3

Clean all accessible system surfaces.

1.4

Disinfect all accessible system surfaces. Ensure that no liquids enter the housing.

A.2) DISINFECTION OF WATER CIRCUITS

1.1

see D) page 5 disinfection of water circuits every 14 days

B) after every operation

B.1) SURFACE DISINFECTION

1.1

Close the CAN-jack with the matching cover.

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Use only pre-soaked, ready-to-use disinfectant tissues.

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A.2) Disinfection of water circuits

1.1

see D) page 5 disinfection of water circuits every 14 days

B) after every operation

B.1) Surface Disinfection

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A) prior to initial operation; prior to storing | B) after every operation

REQUIRED MATERIAL (ready-to-use disinfectant tissues)	
Product name	Manufacturer
Bacillol	Bode
CaviWipe	Metrex
Super Sani Cloth	PDI
Meliseptol HBV	B. Braun
Mikrozid AF	Schülke & Mayr

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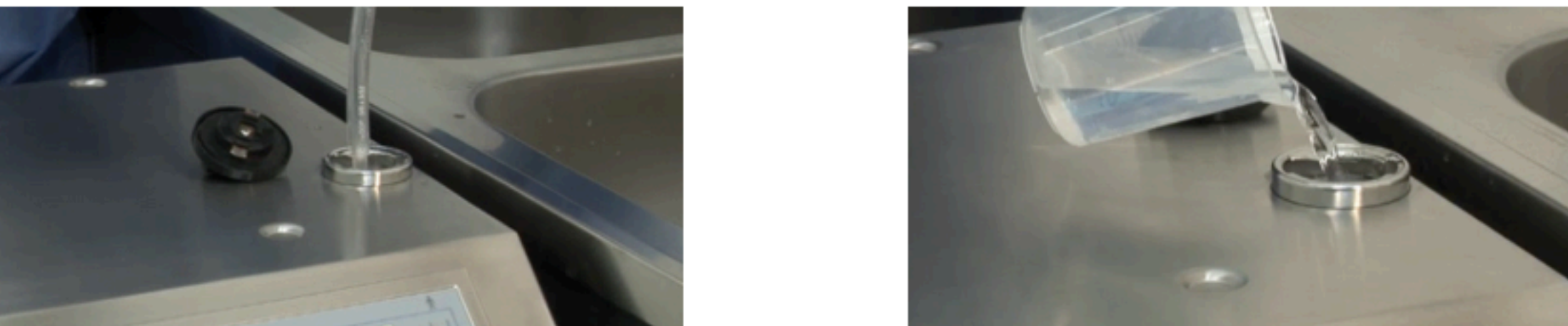
C.1) DRAINAGE

- 1.1 Drain the water tanks.
- 1.2 Be sure that the drain valves are closed once the tanks are empty.



C.2) FILL AND ADD HYDROGENE-PEROXIDE

- 2.1 Open the filling. Fill the device with filtered tap water until the first orange bar is blinking.
- 2.2 Add 150ml hydrogen-peroxide (H₂O₂).



- 2.3 Fill the device with filtered tap water until 2 bars are green. Close tank filling.
- 2.4 Disconnect short circuit tubing patient 1.



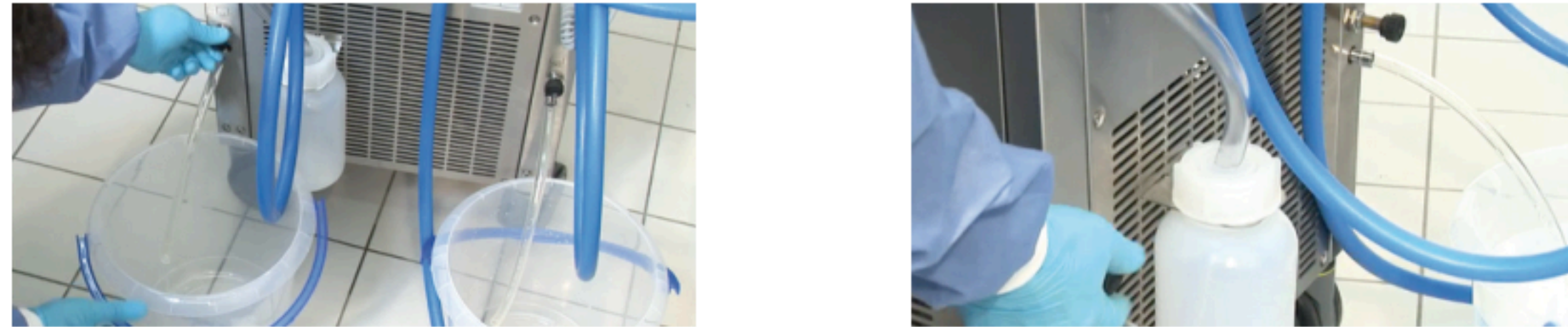
- 2.5 Create a short circuit between inlet of cardioplegia and inlet of the patient 1 while disinfecting the tubing and device connector.
- 2.6 Start cold cardioplegia. Circulate for 5min.



- 2.7 Stop cold cardioplegia.
- 2.8 Switch off the device.

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- 2.7 Stop cold cardioplegia.
- 2.8 Switch off the device.

C.3) DISINFECTION OF OVERFLOW BOTTLE

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3.1 Empty the overflow bottle.



3.2 Disinfect the overflow bottle with automatic chemical-thermal disinfection in a washer-disinfector.



C) every 7 days



C.3) Disinfection of overflow bottle

QUICK START INSTRUCTIONS | Heater-Cooler System 3T

3.1 Empty the overflow bottle.



3.2 Disinfect the overflow bottle with automatic chemical-thermal disinfection in a washer-disinfector.



C) e

D) every 14 days

QUICK START INSTRUCTIONS | Heater-Cooler System 3T

D.1) PREPARATION

1.1 Be sure to wear protective glasses and clothing. Disinfect your hands.



1.2 Use the Aquasafe water filter from Pall to fill the device with filtered tap water. Be sure that the filter is not expired yet.



1.3 Switch on the device. Mute the low level alarm.



1.4 Open the filling. Disinfect the filling and the tubing of the water filter. Fill the device with filtered tap water until the first orange bar is blinking.



D.2) ADD DISINFECTANT

2.1 Measure 450ml Puristeril 340 or 450ml Peresal or 450ml Minncare or 180ml Clorox. Fill the disinfectant into the device.



2.2 Add filtered tap water until 2 bars are green. Close the tank filling. Make sure all circuits are closed.



2.3 Disinfect tubing connectors. Disinfect device connectors.



2.4 Create a short circuit between inlet of cardioplegia and inlet of the patient 1. Make sure all circuits are closed.



D) every 14 days

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D.2) add disinfectant

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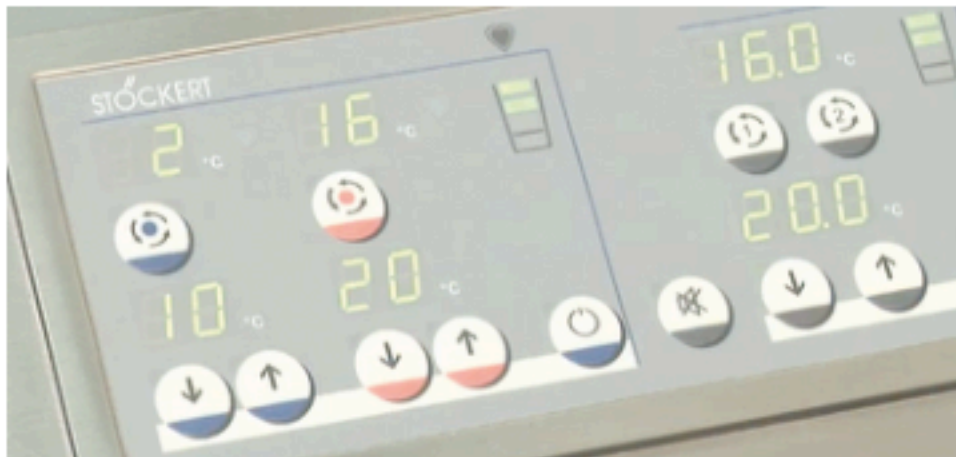
2.3 Disinfect tubing connectors. Disinfect device connectors.



2.4 Create a short circuit between inlet of cardioplegia and inlet of the patient 1. Make sure all circuits are closed.



- 2.5 Set cold cardioplegia to 10°C.
Set warm cardioplegia and patient temperature to 20°C.



D.3) DISINFECT (5 MIN.)

- 3.1 Start cold cardioplegia.
Circulate for 5min.
Stop cold cardioplegia.



- 3.2 Disinfect additional short circuit tubing.
Disinfect device connectors.
Establish a short circuit tubing for patient 2.



- 3.3 Disconnect and disinfect established short circuit.
Disinfect device connector.
Establish short circuit tubing for cardioplegia.



- 3.4 Disinfect connectors of longer tubing and short circuit adaptor. Connect both tubing ending with short circuit adaptor.



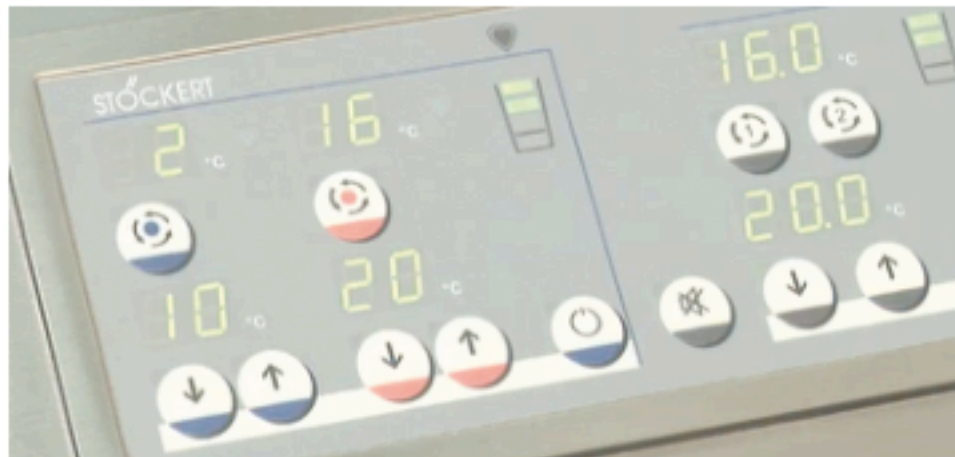
- 3.5 Disinfect device connectors.
Disinfect tubing connectors.
Establish short circuit tubing for patient 1.



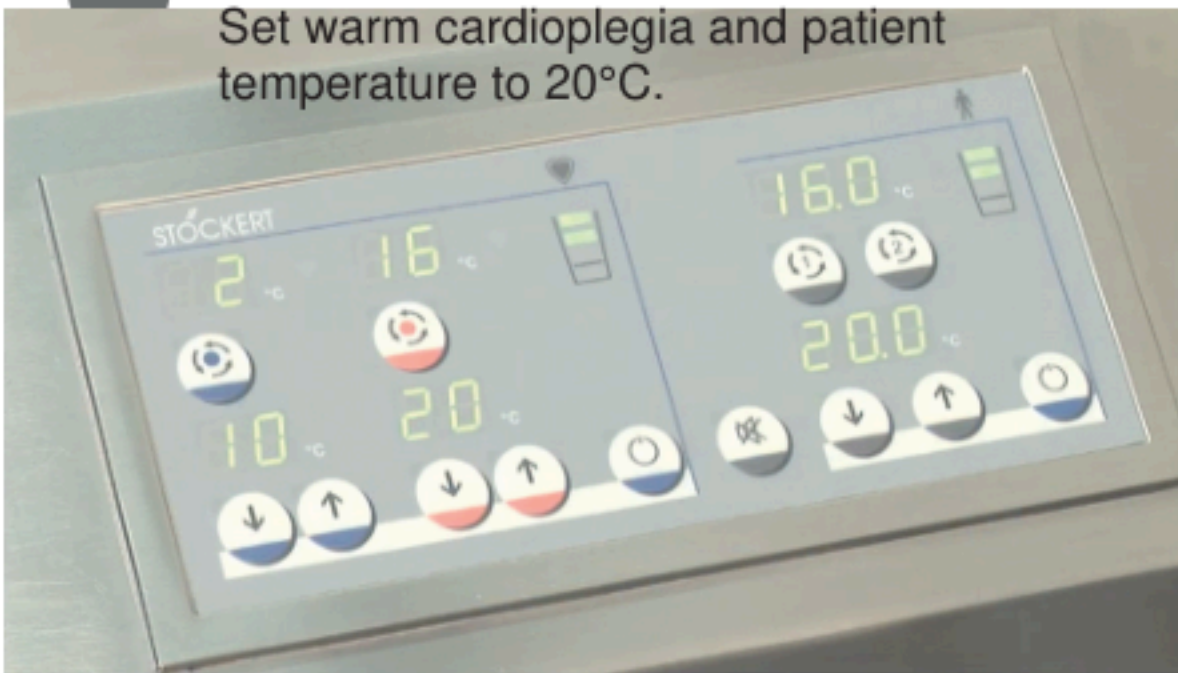
- 3.6 Open all valves.



D) every 14 days



- 2.5 Set cold cardioplegia to 10°C.
Set warm cardioplegia and patient temperature to 20°C.



D.3) disinfect (5 min.)

- 3.1 Start cold cardioplegia.
Circulate for 5min.
Stop cold cardioplegia.



- 3.2 Disinfect additional short circuit tubing.
Disinfect device connectors.
Establish a short circuit tubing for patient 2.



- 3.3 Disconnect and disinfect established short circuit.
Disinfect device connector.
Establish short circuit tubing for cardioplegia.



- 3.4 Disinfect connectors of longer tubing and short circuit adaptor. Connect both tubing ending with short circuit adaptor.



- 3.5 Disinfect device connectors.
Disinfect tubing connectors.
Establish short circuit tubing for patient 1.



- 3.6 Open all valves.



D.4) DISINFECT (10 MIN.)

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- 4.1

Start warm cardioplegia and patient 1 and 2. Circulate for 10min.
- 4.2

Close all valves while circuits are running to empty the tubing.
- 4.3

When tubing are empty stop all circuits.

D.5) DRAINAGE 1

- 5.1

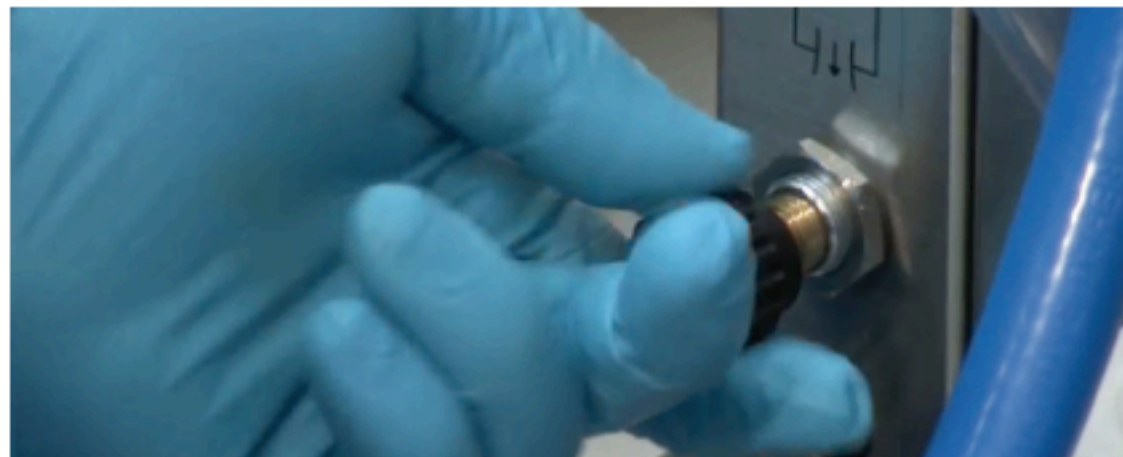
Disinfect drainage tubing and drains. Connect both drainage tubing.
- 5.2

Open both drains. Drain the device completely.



- 5.3

Close both drains.



D) every 14 days

D.4) disinfect (10 min.)

QUICK START INSTRUCTIONS | Heater-Cooler System 3T

- 4.1

Start warm cardioplegia and patient 1 and 2. Circulate for 10min.
- 4.2

Close all valves while circuits are running to empty the tubing.
- 4.3

When tubing are empty stop all circuits.

D.5) drainage 1

- 5.1

Disinfect drainage tubing and drains. Connect both drainage tubing.
- 5.2

Open both drains. Drain the device completely.



- 5.3

Close both drains.



D.6) RINSE 1

- 6.1 Open tank filling.
Fill the device with filtered tap water until 2 bars are green. Close tank filling.



- 6.3 Close all valves while circuits are running to empty the tubing.



D.7) DRAINAGE 2

- 7.1 Open both drains.
Drain the device completely.

D.8) RINSE 2

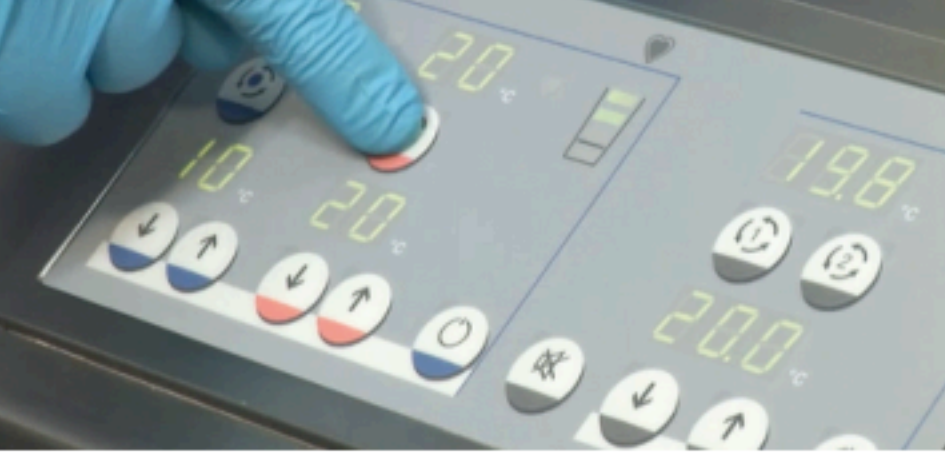
- 8.1 Open tank filling.
Fill the device with filtered tap water until 2 bars are green. Close tank filling.

- 8.3 Close all valves while circuits are running to empty the tubing.

D.9) DRAINAGE 3

- 9.1 Open both drains.
Drain the device completely.

- 6.2 Open all valves.
Start warm cardioplegia and patient 1 and 2. Circulate for 3min.



- 6.4 When tubing are empty stop all circuits.



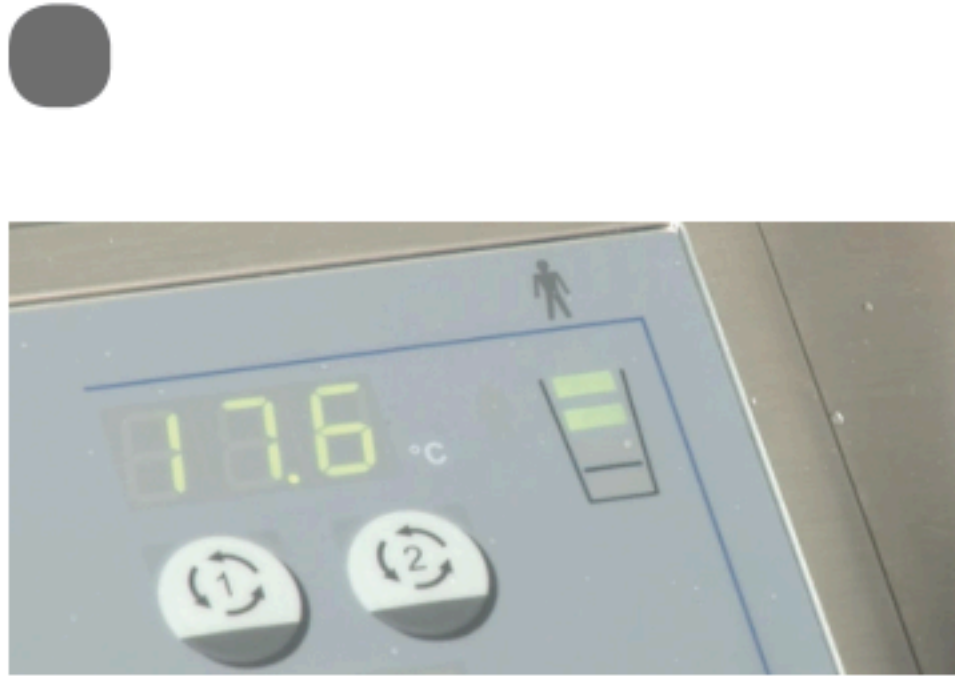
- 7.2 Close both drains.

- 8.2 Open all valves.
Start warm cardioplegia and patient 1 and 2. Circulate for 3min.

- 8.4 When tubing are empty stop all circuits.

- 9.2 Close both drains.

D) every 14 days



D.6) Rinse 1

- 6.1 Open tank filling.
Fill the device with filtered tap water until 2 bars are green. Close tank filling.



- 6.3 Close all valves while circuits are running to empty the tubing.



D.7) Drainage 2

- 7.1 Open both drains.
Drain the device completely.

D.8) Rinse 2

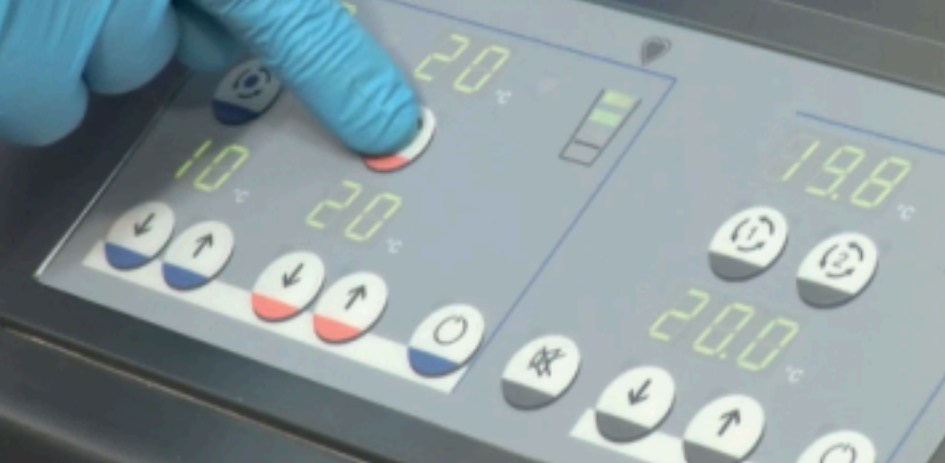
- 8.1 Open tank filling.
Fill the device with filtered tap water until 2 bars are green. Close tank filling.

- 8.3 Close all valves while circuits are running to empty the tubing.

D.9) Drainage 3

- 9.1 Open both drains.
Drain the device completely.

- 6.2 Open all valves.
Start warm cardioplegia and patient 1 and 2. Circulate for 3min.



- 6.4 When tubing are empty stop all circuits.



- 7.2 Close both drains.

- 8.2 Open all valves.
Start warm cardioplegia and patient 1 and 2. Circulate for 3min.

- 8.4 When tubing are empty stop all circuits.

- 9.2 Close both drains.

D.10) FILL AND ADD HYDROGENE-PEROXIDE

QUICK START INSTRUCTIONS | Heater-Cooler System 3T

- 10.1 Open the filling.
Fill the device with filtered tap water until the first orange bar is blinking.



- 10.2 Add 150ml hydrogen-peroxide (H₂O₂).



- 10.3 Fill the device with filtered tap water until 2 bars are green.
Close tank filling.

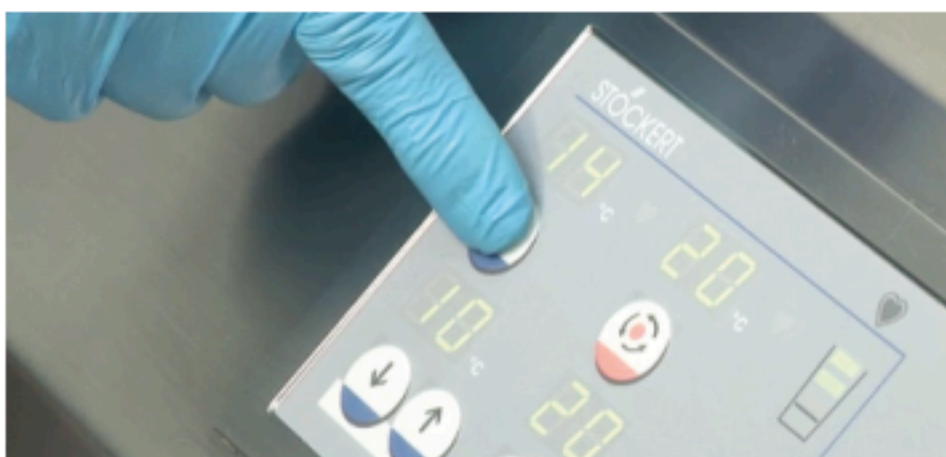


- 10.4 Disconnect short circuit tubing patient 1.

- 10.5 Create a short circuit between inlet of cardioplegia and inlet of the patient 1 while disinfecting the tubing and device connector.



- 10.6 Start cold cardioplegia.
Circulate for 5min.



- 10.7 Stop cold cardioplegia.

- 10.8 Switch off the device.



NOTE: The overflow bottle must also be disinfected regularly (see C.3) page 4).



- 10.7 Stop cold cardioplegia.



D.10) Fill and add hydrogene-peroxide

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Fill the device with filtered tap water until the first orange bar is blinking.



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- 10.4 Disconnect short circuit tubing patient 1.

- 10.5 Create a short circuit between inlet of cardioplegia and inlet of the patient 1 while disinfecting the tubing and device connector.



- 10.6 Start cold cardioplegia.
Circulate for 5min.



- 10.7 Stop cold cardioplegia.

- 10.8 Switch off the device.

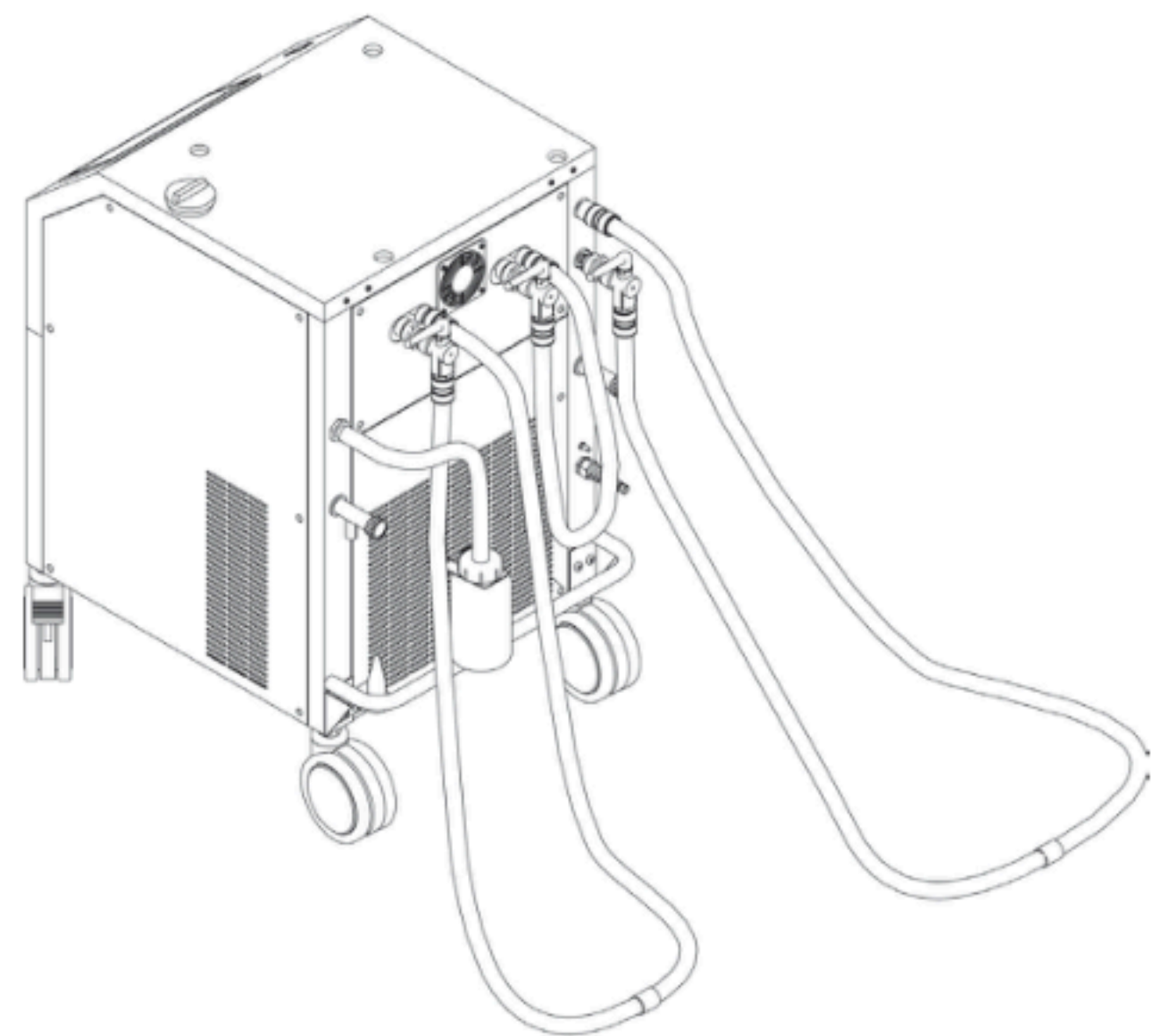


NOTE: The overflow bottle must also be disinfected regularly (see C.3) page 4).

E) once per year

E.1) EXCHANGE THE TUBINGS

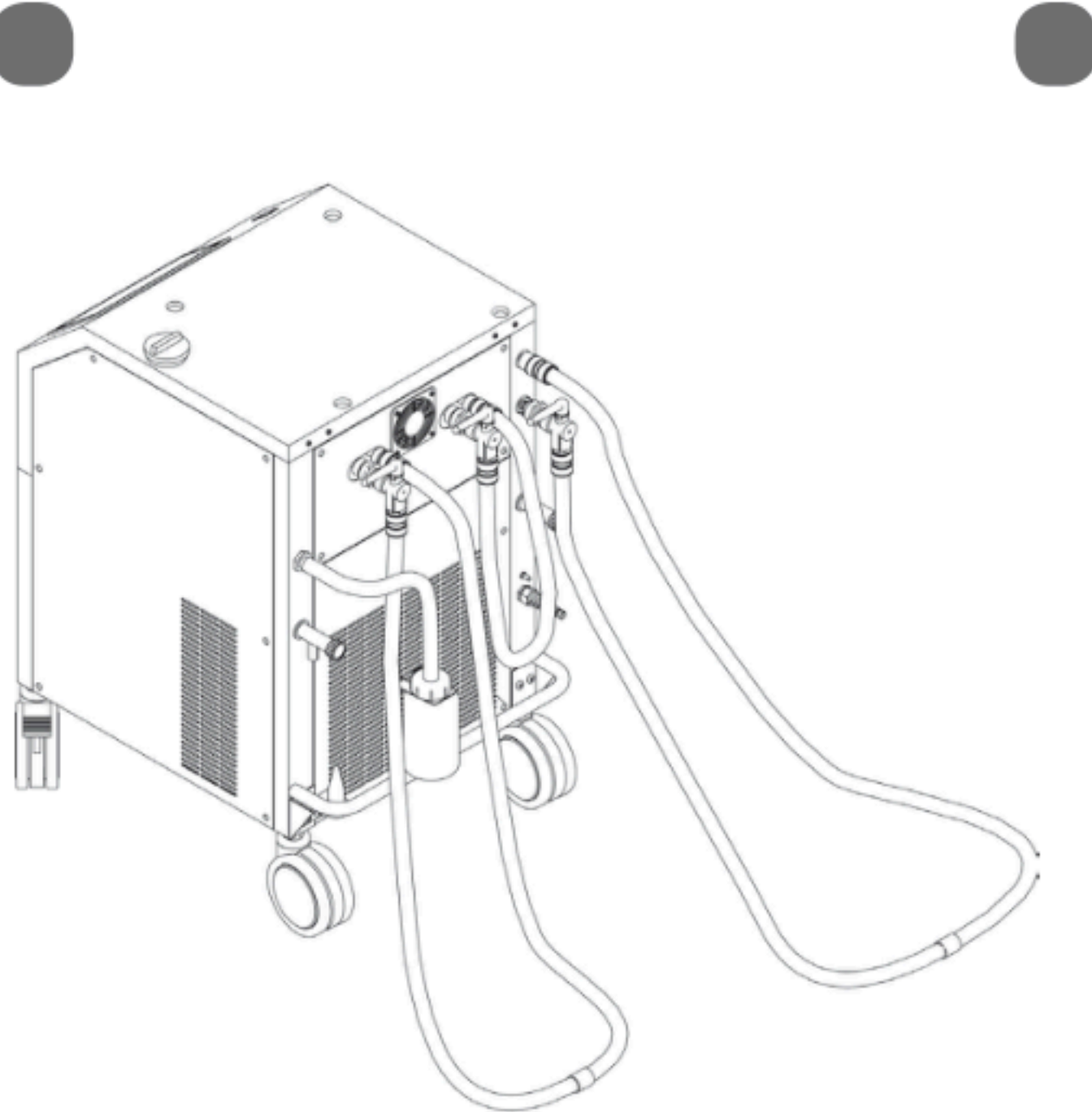
- 1.1 Tubings that are used with the heater-cooler must be replaced once a year.
- 1.2 Include the tubings in the disinfection as described. Connect the tubings during the disinfection cycle to the heater-cooler.



E) once per year | F) regularly

E) once per year

E.1) EXCHANGE THE TUBINGS



F) regularly

F.1) TAKE WATER SAMPLES

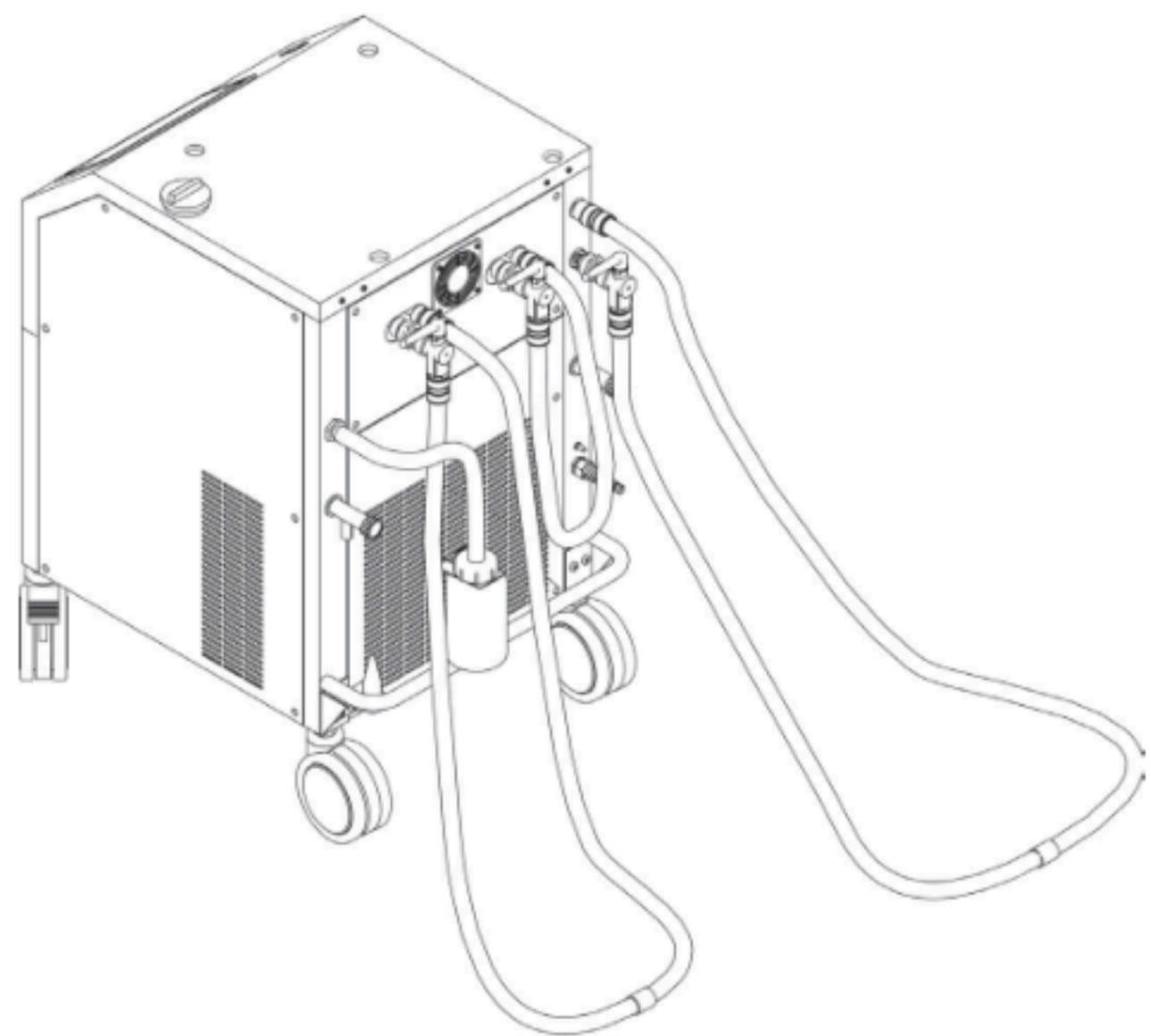
- 1.1 Disinfect the drain ports (spray, wipe).
- 1.2 Only if disinfection has not been performed correctly, water samples shall be taken and analyzed. Open the drain port. Let the water run for a while (5sec) and then take the water sample out of the running water stream.

NOTE: Only use tubings that are certified for drinking water systems, such as SORIN part number 75-510-218.

E) once per year

E.1) Exchange the tubings

- 1.1 Tubings that are used with the heater-cooler must be replaced once a year.
- 1.2 Include the tubings in the disinfection as described. Connect the tubings during the disinfection cycle to the heater-cooler.



F) regularly

F.1) Take water samples

- 1.1 Disinfect the drain ports (spray, wipe).
- 1.2 Only if disinfection has not been performed correctly, water samples shall be taken and analyzed. Open the drain port. Let the water run for a while (5sec) and then take the water sample out of the running water stream.

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